

EE 412/512 Lab #1

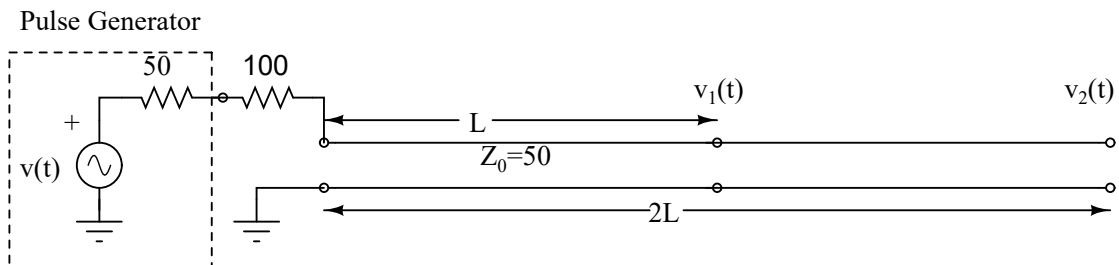
Write the following statement on the first line of your solution sheet:

“I completed this homework assignment independently.”

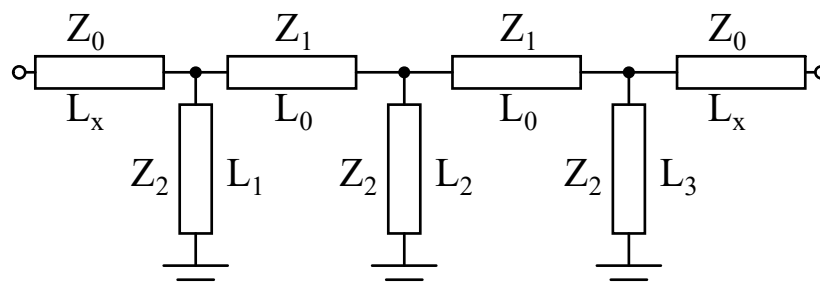
Register and download ADS using Bilkent e-mail and install it on your PC. Read the ADS Tutorial.

Preliminary Work:

- 1) We have a transmission line of length $2L$ ($L=5\text{m}$) excited by a pulse generator as shown below. RG-58 transmission line uses polyethylene as the insulator and hence the velocity factor $v_c/c=0.66$.

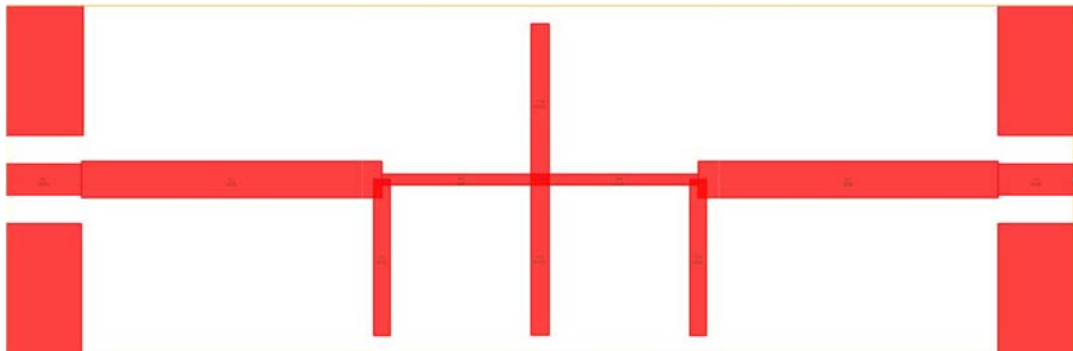
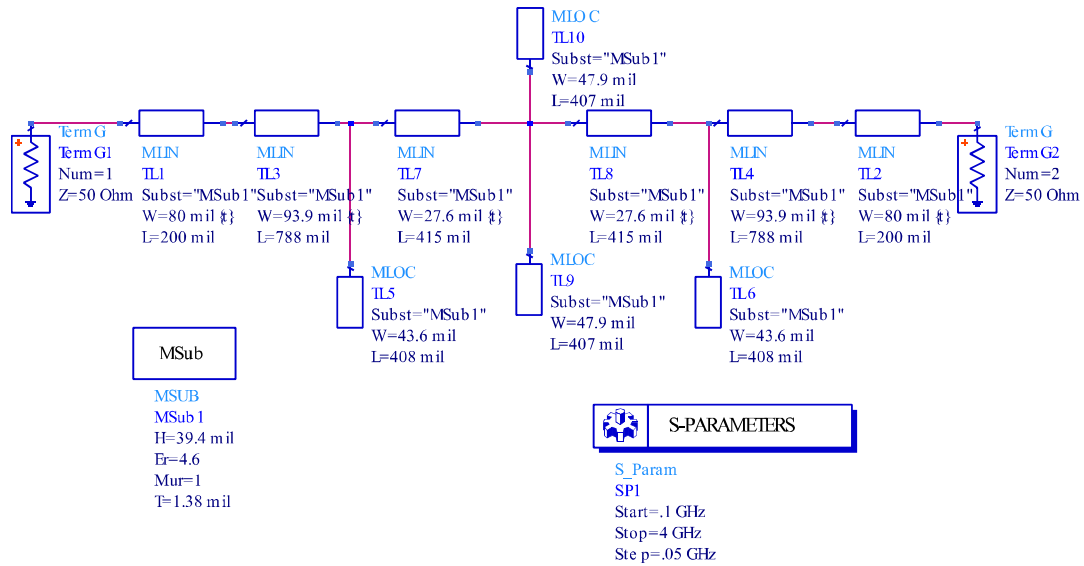


- a) Estimate the voltage at the middle, $v_1(t)$, and end, $v_2(t)$, of the transmission line as a function of time if the generator output $v(t)$ is a step function of amplitude 5V.
 - b) Estimate the same voltages if the generator output $v(t)$ is a pulse of amplitude 5V and of 10ns duration.
- 2) A load of $22\ \Omega$ in series with an inductor of 12 nH is to be fed by a $50\ \Omega$ source. Select one of the frequencies: 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300 MHz. A triple-stub tuner shown below is to be used as the matching network at the chosen frequency. $L_x=50\text{mm}$ and $L_0=88\text{ mm}$ are fixed while L_1 , L_2 , and L_3 are variable between 50 mm to 760 mm. The variable transmission lines (Z_2) use an air insulator ($v_c=c$), while the fixed transmission lines have an insulator such that $v_c/c=0.7$ for Z_1 and $v_c/c=0.8$ for Z_0 . We have $Z_0=50\ \Omega$, $Z_1=36\ \Omega$, and $Z_2=45\ \Omega$. Using ADS, find the lengths of the three shorted transmission lines (using Tuning facility) that can be used for the impedance match. Note that there is no unique solution. Plot the simulated S_{11} in dB (–50 to 0)

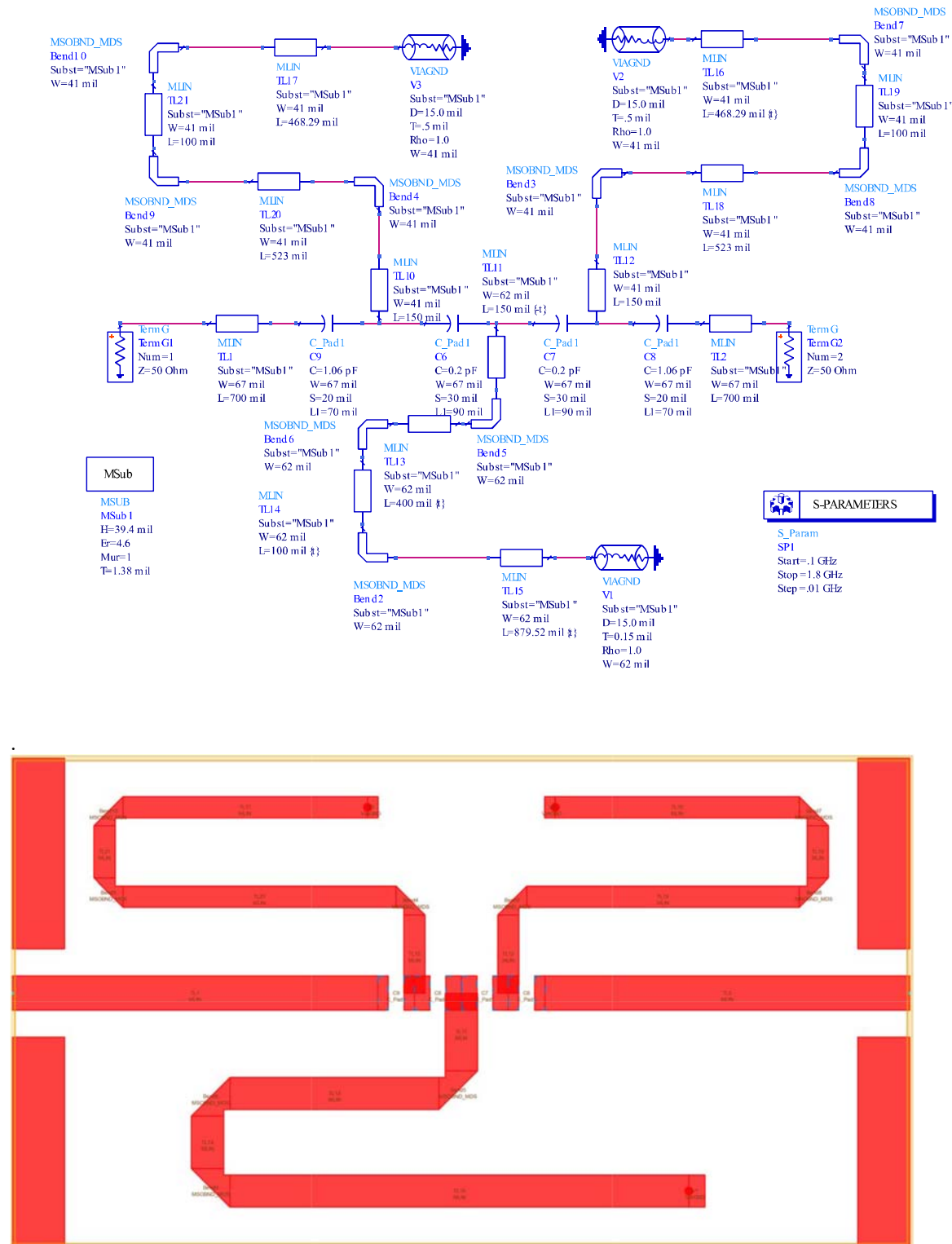


3) We have a microstrip low-pass filter built on a substrate with $H=1\text{mm}$, $\epsilon_r=4.6$, with a copper thickness of $35\mu\text{m}$. The dimensions of the microstrip lines are given in the schematic.

- Enter the schematic in ADS and plot the S_{11} and S_{21} of the filter (in dB from -50 to 0)
- Modify the filter so that the corner frequency is different. Plot the S_{11} and S_{21} of your filter (in dB from -50 to 0). Generate the layout of your filter.



- 4) We have a microstrip band-pass filter built on the same substrate.
 - a) Using the given schematic, plot the S_{11} and S_{21} of the filter (in dB from -50 to 0).
 - b) Generate another band-pass filter with a different center frequency. Plot the S_{11} and S_{21} of the filter (in dB from -50 to 0). Generate the layout of your filter.



Experimental Work:

- 1) Setup the transmission line circuit of Preliminary step 1. Measure the voltages $v_1(t)$ and $v_2(t)$ for
 - a) step input
 - b) pulse input.Compare with the estimate.
- 2) Set the stub lengths of the triple-stub tuner to the calculated values.
 - a) Measure S_{11} (in dB from -50 to 0).
 - b) Optimize the stub lengths for a better match at the chosen frequency. Measure the stub lengths.
- 3) Measure the S_{11} and S_{21} of the fabricated low-pass filter (in dB from -50 to 0). Compare with the simulated values.
- 4) Measure the S_{11} and S_{21} of the fabricated band-pass filter (in dB from -50 to 0). Compare with the simulated values.